Firm Debt Relief in Financial Downturn

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Debt relief policy usually targets firms by size, most commonly small firms. I study if that is the best target of a fixed budget during a financial crisis in a DSGE model of heterogeneous firms, financial frictions, and aggregate uncertainty that reproduces key aspects of U.S. data, including the size and age-size distributions of firms. Optimal policy relieves debt where the excess return to capital investment is highest, cutting the output trough by 47 percent. Small size is a poor proxy for this; targeting young or medium-size firms improves welfare by 5 and 3.5 times that of small firms, respectively.

JEL: E22, E32, E62, H32, C60

Can targeting firm debt relief improve stabilization policy in a financial crisis? In economic downturns, the idea of debt relief, or "bailouts," becomes more common in political and economic discourse.[1] These policies are usually idiosyncratic, varying in size and scope, and may also allow for changes in market structures. In recent U.S. history, debt relief policies have been implemented during the Great Recession and the COVID-19 pandemic. Motivation for policy during the Great Recession was driven by concerns about larger firms, specifically the "Big Three" in the U.S. auto industry, laying off workers and a negative wealth effect spreading throughout the economy.[2] Furthermore, larger firms are responsible for a greater share of aggregate economic activity than smaller firms.[3] On the other hand, the

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[1] Blau, Brough and Thomas (2013) measure commonality in political discourse with two approximations for political engagement: lobbying and connections maintained.

[2] General Motors, Ford Motor Company, and Chrysler, at the time, were considered the "Big Three" U.S. automobile manufacturers.

[3] Crouzet and Mehrotra (2020) conclude that cyclical behavior in large firms may have implications for aggregate fluctuations, while small firms' behavior may not. This, along with the positively skewed employment distribution in the United States shown in the U.S. Census Business Dynamics Statistics (2022) (BDS) data, may suggest targeting large firms is appropriate.

Paycheck Protection Program (PPP) established during the pandemic was chiefly concerned with smaller businesses, with the main target being firms with 500 or fewer employees.[4] The rationale for targeting smaller firms is based on the fact that, on average, they are likely to grow faster than larger firms on a rate basis, despite weaker capital market access.[5] Beyond the question of which end of the size distribution to focus resources on, it remains to be seen if targeting size at all is the best use of finite funds.

To answer the question, I build a dynamic stochastic general equilibrium model with persistent firm-level heterogeneity, financial frictions, and aggregate uncertainty. I find that the optimal policy at the onset of a crisis is one that decreases and equalizes the highest levels of excess return to physical capital investment across firms, while internalizing its own pecuniary externality.[6] Such a policy lessens the fall of both output and investment by nearly half at their troughs, which is equivalent to a 0.093 percent increase in *permanent* consumption for the household.

Many policies in the United States and abroad condition on size and age, with small-firm targets being the most common.[7] However, I find that policy that conditions on size provides the best returns when targeting medium-size firms, not small or large. The medium-size policy makes the fall in output and investment over 10 percent less severe, capturing about 20 percent of the welfare gains from the excess return policy. Moreover, results show that size may not be the closest substitute for an optimal policy at all. Conditioning on age and targeting young firms captures 28 percent of the excess return policy's welfare gains, suggesting that common policy criteria may need to be reconsidered.

The heterogeneity in production creates differences in optimal investment decision rules, while financial frictions can constrain some of these decisions by limiting firms' access to debt markets. The result is a dynamic distribution of firms, which provides a breadth of

[4]The second draw targeted firms with 300 or fewer employees. Other financial targets were also implemented, such as net-worth limits and the exclusion of hedge funds.

[5]Evans (1987) and Hall (1987) provide empirical work supporting this claim. Banz (1981) finds that firm size is negatively correlated with risk-adjusted returns; much work has been done discussing this correlation and whether or not causality exists (Berk, 1997; Asness et al., 2018). Gertler and Gilchrist (1994) state, "While size per se may not be a direct determinant, it is strongly correlated with the primitive factors that do matter," suggesting policy would be better off targeting smaller firms.

[6]Excess return to capital investment is defined as the expected lifetime marginal value of investing in capital today, minus its cost. See Appendix A3 for an approximation of the externality.

[7]Congressional Budget Office (2020), Jo and Senga (2019), Guner, Ventura and Xu (2008).

firm-level variables to target for policy analysis. The aggregate uncertainty in the model is in two dimensions: credit conditions and aggregate productivity. This is a critical feature as the central object that a targeted policy hopes to condition on is the shadow value of a unit of liquidity to a firm. This is a forward-looking expectation that depends on whether aggregate conditions persist, and so uncertainty directly impacts the efficacy of policy. Further, policymakers assuming they have perfect foresight may misjudge which firms that dollar is worth most to.[8]

I model financial market frictions by assuming collateralized borrowing. Investment among firms with insufficient collateral is limited by their inability to borrow, leading to a misallocation of capital. A dynamic fraction of the distribution of firms may not yet have the required capital stock to post sufficient collateral, nor the required retained earnings to reach their efficient investment level, slowing their growth. A shock to firms' access to credit exacerbates the misallocation already present in the model, leading to a deterioration of the aggregate capital stock. The recovery path is a function of the distribution of firms, which is inherently a slow-moving object, and if capital requires "time-to-build," the recovery is slowed further, as firms are not able to update their capital immediately. Critically, firms still want to grow, as nothing has made the economy exogenously less productive, but the lack of credit misallocates those productive resources.

The model is calibrated to match key moments in U.S. economic data, including the heavily skewed unconditional size distribution of firms. Further, the model produces a monotonically increasing age-size distribution for ages 1–5, which is untargeted; only the average for ages 0–5 enters the calibration. This paper is unique in that, to my knowledge, it is the first of its class that produces both distributions. Moreover, the model is over-identified, serving as further validation for the structure and calibration.

The paper uses the solution to the government's problem as a reference for comparison and to theoretically motivate results. Young firms face high levels of excess return, as it takes time to outgrow financing constraints. Though small firms have the least collateral to secure their loans, they also have the lowest efficient investment target, contrasting with

[8]Holmström and Tirole (1998) illustrate the benefits of public liquidity under aggregate uncertainty as the private supply will be insufficient.

large firms that have the highest self-financing capacity, but the highest investment target. Medium-size firms operate in the middle, lacking the financing capacity of large firms, while having a greater investment demand than small firms. As such, they face the highest excess return on average, and policy has more to correct.

The benefits of policy are not without costs. The household pays a lump-sum tax to fund policy; it forgoes present consumption at the onset of the crisis for a potentially improved recovery path. Since all prices are a function of the marginal utility of consumption, this merits specific consideration as the model is solved non-linearly and equilibrium decision rules are re-solved for all firms at each date. The shape of the endogenously evolving distribution matters for both the direct response to an injection of resources and the general equilibrium effects of price changes, especially when the shock is not proportional across the distribution, and prices are being forecasted by the household with uncertainty over future states.

The remainder of the paper is organized as follows. Section I provides a review of the literature. Section II describes the model and provides analysis. Section III begins the quantitative exercise with the calibration. Section IV quantifies measures of inefficiency in the model, while Section V discusses policy to remedy the effects of a financial crisis. Section VI discusses welfare results, and Section VII concludes.

I. Literature Review

There is a large literature studying the effects of fiscal policy on financial variables. Jo and Senga (2019) and Buera, Moll and Shin (2013) study the long-term effects of policy intended to lessen the effects of financial frictions over a distribution of producers. Bianchi (2016) studies debt relief in a crisis, and Elenev, Landvoigt and Van Nieuwerburgh (2022) look at bridge loans during COVID-19-motivated shocks with financial consequences. I contribute to this literature by deriving an optimal policy for a non-degenerate distribution of firms that takes into account not only which firms are eligible, but how much each specific firm should receive. Moreover, I analyze policy effects in a crisis where the distribution is endogenous, instead of long-run comparisons or transitions with degenerate distributions. This allows for a better understanding of the firm-level impact during periods where governments are most likely to implement policy. Specifically, in Bianchi (2016), the result against targeted relief

is born out of anticipation of policy, which increases borrowing ex ante. The policy here is unanticipated (both the tax and relief), which closes that channel to isolate the question of whom to target and why. Finally, this structure allows for general equilibrium effects through endogenous interest rate changes, something Jo and Senga (2019), Buera, Moll and Shin (2013), and Bianchi (2016) abstract from.

A growing body of empirical work is highlighting the macroeconomic importance of young firms over small firms. Both Haltiwanger, Jarmin and Miranda (2013) and Kim et al. (2024) show that young firms are more likely to have higher growth rates and that, controlling for age, the relationship between size and growth is weak. My work applies a well-studied theoretical structure to illustrate a potential mechanism, delivered through a measurement of each individual firm's marginal valuation of investment, excess return. Moreover, this doubles as the main variable for optimal policy. Further, Dinlersoz et al. (2018) document a positive relationship between firm leverage and size, but a negative relationship with age, which the model reproduces. They conclude that the Great Recession disproportionally hurt firms that rely on credit access, young firms trying to pay down their leverage, and not the highly levered large firms that have cash reserves.[9] These empirical results highlight the importance of not only matching the distributions in the data, but also matching the trend in their decisions in order to understand what policies may work best in a financial crisis.

Such a structure is built on work modeling financial frictions from Kiyotaki and Moore (1997), Jermann and Quadrini (2012), and Khan and Thomas (2013).[10] Imperfect access to credit markets directly limits investment and thus the rate of firm growth. The interaction of these frictions with persistent firm-level and aggregate uncertainty creates decision rules for capital and debt that vary across firms. Cooley and Quadrini (2001) document how size and age dynamics can arise from these interactions. My quantitative strategy maps that to both the size and age-size distributions of firms in the United States in order to better understand the impacts of commonly used policies aimed at relieving these financial frictions.

In this work, the government's role is akin to that in Woodford (1990): it provides a public

[9]This contrasts with the canonical "hump-shaped" leverage measure seen in public firms only (Faff et al., 2016). These firms are usually years into their life cycle when they enter Compustat.

[10]See Albuquerque and Hopenhayn (2004) for results driven by limited enforceability and Bernanke and Gertler (1989) for agency costs.

avenue to circumvent financial constraints. This is expanded on by Holmström and Tirole (1998) and Angeletos, Collard and Dellas (2023) with additional micro-foundations for liquidity constraints and consideration for policy costs. Further, Dávila and Korinek (2018) show that price changes in these settings are not welfare neutral, introducing another potential policy cost. Optimal policy in this paper is built on quantifying the inefficiencies from the micro-foundations, as well as providing an approximation of the pecuniary externality.

II. Model

A. *Firms*

The model economy is populated with a unit mass of firms that use labor, n, and predetermined capital, k, to produce a homogeneous consumption good through a decreasing-returns-to-scale, twice-differentiable production function, $y = z\varepsilon F(k, n)$. The variable z is aggregate, exogenous total factor productivity (TFP), and ε is the idiosyncratic counterpart. Both are stochastic; I assume both z and ε follow their own Markov processes: $z \in \mathbf{Z} \equiv \{z_1, \ldots, z_{N_z}\}$, where $\Pr(z' = z_g \mid z = z_f) = \pi^z_{f,g}$ and $\varepsilon \in \mathbf{E} \equiv \{\varepsilon_1, \ldots, \varepsilon_{N_\varepsilon}\}$, where $\Pr(\varepsilon' = \varepsilon_j \mid \varepsilon = \varepsilon_i) = \pi^\varepsilon_{i,j}$. Both $\sum_{g=1}^{N_z} \pi^z_{f,g} = 1$ and $\sum_{j=1}^{N_\varepsilon} \pi^\varepsilon_{i,j} = 1$. I assume ε is drawn from a bounded Pareto distribution with bounds ε_L and ε_H, and shape η_P. Each period, there is a ρ_ε probability that firms keep their current ε. With probability $(1 - \rho_\varepsilon)$, they draw a new value from the ergodic distribution implied by $\pi^\varepsilon_{i,j}$, $H(\varepsilon)$.

In order to fund investment, i, for next period's capital, k', firms use their revenues and issue a risk-free bond, b', and b represents their current debt.[11] They face collateralized borrowing constraints and may not borrow more than a given fraction, ζ, of their current capital stock each period. Collateralized borrowing is a reduced-form method of capturing the health of financial markets. A higher ζ reflects lenders' positive beliefs in the market; thus, they are willing to lend more to firms. On the other hand, a lower ζ reflects just the opposite. The parameter $\zeta \in \boldsymbol{\zeta} \equiv \{\zeta_1, \ldots, \zeta_{N_\zeta}\}$ is stochastic and follows its own Markov process, where $\Pr(\zeta' = \zeta_k \mid \zeta = \zeta_h) = \pi^\zeta_{h,k}$ and $\sum_{k=1}^{N_\zeta} \pi^\zeta_{h,k} = 1$.

Aside from collateral constraints, firms also face a lower bound on dividends. The dividend restriction gives the collateral constraint its "bite." If a firm cannot afford to finance

[11] The convention used is that $b > 0$ represents debt and $b < 0$ represents net savings.

investment and it cannot provide enough collateral for a loan, this firm may not make up the difference by issuing sufficiently negative dividends.[12] This allows the model to capture some of the effects described in the corporate finance literature on the preference for debt financing over equity financing in a tractable manner, while still focusing a quantitatively robust analysis on debt channels.[13]

As firms age, their age being denoted by a, their borrowing capacity increases with capital accumulation. In order to prevent the whole economy from outgrowing financial conditions, each period a fraction, $\pi_d(a)$, of firms exogenously exit the market permanently at the end of a period. To reflect the negative correlation between age and exit rates seen in the U.S. Census Business Dynamics Statistics (2022) (BDS), $\pi_d(a)$ is a function of firm age. Thus, a given firm, (k, b, ε, a), is defined by its current capital, $k \in \mathbf{K} \subset \mathbf{R}_+$, debt level, $b \in \mathbf{B} \subset \mathbf{R}$, idiosyncratic shock realization, and age, $a \in \mathbf{A} \subset \mathbf{Z}_+$, with an absorbing state $a = a_{\max}$ to close the dimension. An exiting firm is replaced by an entrant with capital stock k_0, debt b_0, and idiosyncratic productivity, ε_0, from $H(\varepsilon)$, drawn from an infinite pool of potential entrants to ensure the mass of firms will not change. The resulting entry rate is π_e.

At the start of the period, firms are made aware of their idiosyncratic, ε, and aggregate productivity, z, the financial state, ζ, and if they will exit the market. There is no penalty for exiting the market; firms know if they will exit and, if so, optimize their labor choice given their current capital and productivity, produce output, and settle any outstanding financial obligations, which had an upper bound from last period's collateral constraint. They do not borrow or invest. This timing is crucial, as it ensures all debt is risk-free.

Given the structure of the production sector, I now characterize the state of the economy. The distribution of firms over (k, b, ε, a) is described by the probability measure μ defined on the Borel algebra, $\mathcal{S}$, generated by the open subsets of the product space, $\mathbf{S} = \mathbf{K} \times \mathbf{B} \times \mathbf{E} \times \mathbf{A}$. The distribution of firms evolves over time according to the mapping, Γ, from the current aggregate state, $\mu' = \Gamma(z, \zeta, \mu)$. For expositional ease, let the exogenous components of the aggregate state be $s = (z, \zeta)$ and its transition probabilities be $\pi^s_{l,m} = \Pr(s' = (z, \zeta)_m \mid$

[12] See Bianchi (2016) as an example of setting a lower bound on dividends as a way to balance tractability with quantitative robustness.

[13] Shleifer and Vishny's 1997 research survey and Brav et al.'s 2005 managerial survey show that some firms will choose external capital or even give up positive-NPV investments in order to not decrease dividends.

$s = (z, \zeta)_l)$, where each $\pi^s_{l,m}$ is derived from the transition probabilities $\pi^{\zeta}_{h,k}$ and $\pi^{z}_{f,g}$. Let the shocks realized in the aggregate state s_l be z_f and ζ_h, for each $l \in \{1, \ldots, N_s\}$, where $N_s = N_z N_\zeta$. Finally, let S represent the entire aggregate state, $S_l = (s_l, \mu)$.[14]

The Firms' Problem

With the aggregate state described, I now begin to characterize a given firm's problem. Its cash-on-hand is its output, less wage bill, plus undepreciated capital stock, minus debt obligations, with the potential share of debt relieved denoted as $g(k, b, \varepsilon, a; S)$. Capital depreciates at rate δ and takes one period to build. Therefore, cash-on-hand is written as:

$$x(k, b, \varepsilon, a; S) = z\varepsilon F(k, n) - w(S)n + (1 - \delta)k - (1 - g(k, b, \varepsilon, a; S))b \tag{1}$$

and $(1-g(k, b, \varepsilon, a; S))$ is the fraction of debt the firm is responsible for. This policy structure, where the government retires a fraction of firm debt, is akin to that of Bianchi (2016).

Before making any decisions, a firm's value is then given by $v_0(k, b, \varepsilon, a; S)$, the probability-weighted sum of exiting with cash-on-hand or continuing on to the next period and receiving continuation value, $v(k, b, \varepsilon, a; S)$:

$$v_0(k, b, \varepsilon_i, a; S) = \pi_d(a)x(k, b, \varepsilon_i, a; S) + (1 - \pi_d(a))v(k, b, \varepsilon_i, a; S) \tag{2}$$

Continuing firms choose dividends, D, borrowing or saving, b', at the loan discount factor $q(S)$, and capital for next period, k', while not exceeding their collateral constraint of $b' \leq \zeta k$. Capital accumulation is $k' = i + (1 - \delta)k$, and $d_m(S_l)$ is the relative valuation of dividends next period, conditional on the current and expected future state. Given the transition conditions of the aggregate and idiosyncratic states, continuation value is:

$$v(k, b, \varepsilon_i, a; S_l) = \max_{n,k',b',D} \left[D + \sum_{m=1}^{N_s} \pi^s_{l,m} d_m(S_l) \sum_{j=1}^{N_\varepsilon} \pi^{\varepsilon}_{i,j} v_0(k', b', \varepsilon_j, a + 1; S'_m) \right] \tag{3}$$

subject to:

[14] Hereafter, I suppress the indices for the current exogenous aggregate state, l, and idiosyncratic productivity, i, except where necessary.

$$
\begin{aligned}
&0 \leq D \leq x(k, b, \varepsilon_i, a; S_l) + q(S_l)b' - k' \\
&b' \leq \zeta_h k \\
&\mu' = \Gamma(S_l)
\end{aligned} \tag{4}
$$

There are no intertemporal factors of labor demand, and the wage bill is paid out of production, meaning all firms with the same (k, ε) have the same labor demand. Thus, the firm's labor decision rule is $g_N(k, \varepsilon; S)$. Similarly, a firm's output is $y(k, \varepsilon; S)$. Given that $\pi_d(a)$ varies with age, and with it the potential valuation of future objects by firms, and that debt relief can condition on age, the capital, debt, and dividend decision rules require arguments from the full individual state; they are $g_K(k, b, \varepsilon, a; S)$, $g_B(k, b, \varepsilon, a; S)$, and $g_D(k, b, \varepsilon, a; S)$, respectively.

B. Household

There is a representative household endowed with ownership of the firms and a unit of time that may be divided between labor and leisure. The household supplies labor to the firms in exchange for a wage, w, transfers resources intertemporally by purchasing bonds, κ', and shares, $\lambda^{h\prime}$, and chooses consumption, c, to maximize its lifetime value, $W(\lambda^h, \kappa; S)$. The discount factor is $\beta \in (0, 1)$. The household maximizes its period utility over consumption and leisure, $U(c, 1 - n^h)$, subject to its budget constraint:

$$
W(\lambda^h, \kappa; S_l) = \max_{c, n^h, \lambda^{h\prime}, \kappa'} \left[U(c, 1 - n^h) + \beta \sum_{m=1}^{N_s} \pi^s_{l,m} W(\lambda^{h\prime}, \kappa'; S'_m) \right] \tag{5}
$$

subject to:

$$
\begin{aligned}
&c + q(S_l)\kappa' + \int \rho_1(k', b', \varepsilon', a'; S_l)\lambda^{h\prime}(d[k' \times b' \times \varepsilon' \times a']) + T \leq \\
&w(S_l)n^h + \kappa + \int \rho_0(k, b, \varepsilon, a; S_l)\lambda^h(d[k \times b \times \varepsilon \times a]) \\
&\mu' = \Gamma(S_l)
\end{aligned} \tag{6}
$$

where $\rho_1(k', b', \varepsilon', a'; S_l)$ is the ex-dividend price of a share, $\rho_0(k, b, \varepsilon, a; S_l)$ is the dividend-inclusive value of a share, and T is lump-sum taxes. The household's decision rules are: $h^C(\lambda^h, \kappa; S)$ for consumption, $h^N(\lambda^h, \kappa; S)$ for labor hours, $h^\kappa(\lambda^h, \kappa; S)$ for bonds, and $h^\lambda(k', b', \varepsilon', a', \lambda^h, \kappa; S)$ for shares in firms with next-period values of k' capital, b' debt, and ε' idiosyncratic productivity, of age a'.[15]

C. *Government*

The role of the government is to conduct debt relief in a crisis. However, the government cannot generate these resources out of nothing; a lump-sum tax is levied on the household in order to fund policy. The benefits of the policy are dependent on resources reaching firms that are financially constrained and face a high marginal productivity of capital to sufficiently compensate the household for the lost consumption. Thus, the government's problem is to maximize household welfare, $\mathcal{W}$, by choice of debt relief targets, subject to its budget, T, the firm-level information that is available to it, $\mathcal{I}$, and the path of prices, $\mathcal{P}$. Let the aggregate state at the relief date be denoted as $S_1 = (s_1, \mu_1)$ so that the formal problem is:

$$\max_{g} \ \mathcal{W}\big(g; \mathcal{P}\big) \tag{7}$$

subject to:

$$T = \int_{\mathbf{S}} g(k, b, \varepsilon, a; S_1) b \mu_1(d[k \times b \times \varepsilon \times a]) \tag{8}$$

$$g(k, b, \varepsilon, a; S_1) \in \big[0, \ \mathbb{1}\{b > 0\}\big] \tag{9}$$

$$g \text{ is } \mathcal{I}\text{-measurable} \tag{10}$$

$$\mathcal{P} \text{ taken as given} \tag{11}$$

Constraint (9) imposes that this policy must be debt relief. The government cannot target firms that are net savers and cannot relieve more than 100 percent of firm debt. Finally, constraint (11) ensures that there is a closed-form solution. This is discussed further in

[15]The household can choose shares of these firm types in equilibrium, since it knows the transition probabilities of the aggregate and idiosyncratic states and the law of large numbers applies. Further, it has access to a complete set of state-contingent claims. These assets are in zero net supply in equilibrium; for simplicity I do not model them here.

Section V.A and Appendix A3, which approximates the difference in welfare between taking $\mathcal{P}$ as given and the full solution to be less than 0.00005 percentage points.

D. Recursive Equilibrium

A *recursive competitive equilibrium* is a set of functions that solve the firms' and household's problems and clear the asset, labor, and output markets:

$$\left(w, q, \{d_m\}_{m=1}^{N_s}, \rho_0, \rho_1, v_0, g_N, g_K, g_B, g_D, W, h^C, h^N, h^\lambda, h^\kappa\right)$$

(i) v_0 solves (2-3) subject to (4), g_N is the exiting firms' decision rule, and (g_N, g_K, g_B, g_D) are the decision rules for continuing firms

(ii) W solves (5) subject to (6), and $(h^C, h^N, h^\lambda, h^\kappa)$ are the household's decision rules

(iii) Share market clearing:
$h^\lambda(k', b', \varepsilon_j, a', \lambda^h, \kappa; S) = \mu'(k', b', \varepsilon_j, a'; S)$ for $(k', b', \varepsilon_j, a') \in \mathbf{S}$

(iv) Labor market clearing: $h^N(\lambda^h, \kappa; S) = \int_{\mathbf{S}} [g_N(k, \varepsilon; S)] \mu(d[k \times b \times \varepsilon \times a])$

(v) Goods market clearing: $h^C(\lambda^h, \kappa; S) = \int_{\mathbf{S}} \Big[z\varepsilon F(k, g_N(k, \varepsilon; S))$
$-(1 - \pi_d(a))\big(g_K(k, b, \varepsilon, a; S) - (1-\delta)k\big) + \pi_d(a)(1-\delta)k - \pi_e k_0 \Big] \mu(d[k \times b \times \varepsilon \times a])$

(vi) Bond market clearing via Walras's Law: $h^\kappa(\lambda^h, \kappa; S) =$
$\int_{\mathbf{S}} (1 - \pi_d(a))\, g_B(k, b, \varepsilon, a; S)\, \mu(d[k \times b \times \varepsilon \times a]) + \pi_e b_0$

(vii) For all Borel sets $A \subseteq \mathbf{K} \times \mathbf{B}$, $\varepsilon_j \in \mathbf{E}$, and ages $a' \geq 1$, continuing firms satisfy:

$$\mu'(A, \varepsilon_j, a') = \int_{\mathcal{M}(A, a')} [1 - \pi_d(a)]\, \pi^{\varepsilon}_{i,j}\, \mu(d[k \times b \times \varepsilon_i \times a])$$

where $\mathcal{M}(A, a') \equiv \{(k, b, \varepsilon, a) \mid (g_K(k, b, \varepsilon_i, a; S),\ g_B(k, b, \varepsilon_i, a; S)) \in A,\ \min\{a + 1, a_{\max}\} = a'\}$, and entrants are age 0: $\mu'(A, \varepsilon_j, 0) = \pi_e\, \mathbb{1}\{(k_0, b_0) \in A\}\, H(\varepsilon_j)$

E. Prices

I begin my analysis by deriving a set of optimality conditions from the household's problem that may be subsumed into the firms' problem. Let C and N represent the optimal

household choices of consumption and labor hours, respectively, with C' and N' being the optimal choices next period, when the aggregate state is S'. Then the real wage (12) is the marginal rate of substitution between leisure and consumption. Since the household owns the firms, the firms' stochastic discount factor (13) is the household's marginal rate of substitution between consumption across states. The loan discount factor (14) is the conditional expectation of the marginal rate of substitution between consumption across states.[16]

$$w(S) = \frac{D_2 U(C, 1-N)}{D_1 U(C, 1-N)} \tag{12}$$

$$d_m(S) = \beta \frac{D_1 U\left(C'_m, 1-N'_m\right)}{D_1 U(C, 1-N)} \tag{13}$$

$$q(S) = \beta \sum_{m=1}^{N_s} \pi^s_{l,m} \frac{D_1 U(C'_m, 1-N'_m)}{D_1 U(C, 1-N)} \tag{14}$$

Let $p(S) \equiv D_1 U(C, 1-N)$ be the household's marginal valuation of output. Since the household owns the firms, I multiply the firms' value (3) by $p(S)$ to price all output in units of marginal utility without altering decision rules. Letting $V_0 = p(S)v_0$ and $V = p(S)v$, I rewrite the firms' problem (3), subject to (4), as:

$$V(k, b, \varepsilon_i, a; S_l) = \max_{k', b', D} \left[p(S_l) D + \beta \sum_{m=1}^{N_s} \pi^s_{l,m} \sum_{j=1}^{N_\varepsilon} \pi^\varepsilon_{i,j} V_0(k', b', \varepsilon_j, a+1; S'_m) \right] \tag{15}$$

F. Allocations

A firm's labor choice is static. It chooses $n^*(k, \varepsilon)$ to solve $z\varepsilon D_2 F(k, n) = w$. Output is then $y(k, \varepsilon) = z\varepsilon F(k, n^*(k, \varepsilon))$, which can be used to define cash-on-hand in (1). Turning from static to dynamic decisions, I divide the analysis into three groups of firms: unconstrained, constrained, and at-risk firms.

Unconstrained firms can reach their optimal capital level unimpeded by *any* financial

[16] The risk-free interest rate on bonds is: $r = \frac{1}{q(S)} - 1$.

constraint, including future constraints. An unconstrained firm chooses $k'(k,b,\varepsilon,a) = \text{argmax}_{k'} V(k,b,\varepsilon,a;S)$. For these firms, their k' choice depends only on their productivity and so it is useful to introduce $k^{\text{f}}(\varepsilon)$, which is the level of capital that any firm would choose in a world where financial constraints did not exist. If $k^*(k,b,\varepsilon,a)$ is the optimal choice of capital for any firm in the model, then for unconstrained firms: $k^*(k,b,\varepsilon,a) = k^{\text{f}}(\varepsilon)$. Taking first-order conditions and applying the Benveniste and Scheinkman (1979) condition, the marginal value from investing is:

$$\frac{\beta}{p(S)} \sum_{m=1}^{N_s} \pi^s_{l,m}\, p(S'_m) \sum_{j=1}^{N_\varepsilon} \pi^\varepsilon_{i,j} \Big(z_m \varepsilon_j D_1 F\big(k^{\text{f}}(\varepsilon), n^*(k^{\text{f}}(\varepsilon), \varepsilon_j)\big) + 1 - \delta \Big) = 1 \tag{16}$$

These firms have no excess return to capital investment; debt relief will produce no extra value. Another way to see this is that neither the constraint nor debt enters into this condition. Relief will change neither labor nor capital, and output will remain the same, as the expected marginal discounted value of an extra unit of capital equals its unit price, 1. Their choice of debt is then $b' \in \left[\frac{D+k^{\text{f}}(\varepsilon)-x(k,b,\varepsilon,a;S)}{q(S)}, \zeta k\right]$, to be discussed further below.

Constrained firms are firms currently bound by the collateral constraint. I take advantage of the monotonically increasing value function in k to simplify the problem. Firms will borrow until $b' = \zeta k$ and invest such that:

$$k'(k,b,\varepsilon,a) = x(k,b,\varepsilon,a;S) + q(S)\zeta k - D < k^*(k,b,\varepsilon,a) \tag{17}$$

The strict inequality is the center of the misallocation problem that policy would seek to correct. These firms cannot reach their desired capital level and are unable to hire as much labor or produce as much output next period, compared to an environment without these financial frictions. Moreover, given decreasing-returns-to-scale production, these firms have a higher shadow price on their collateral and thus a higher valuation for liquidity than firms without any binding constraints. Again using the properties of the value function where

some constraints bind:

$$(18)\quad \frac{\beta}{p(S)}\sum_{m=1}^{N_s}\pi^s_{l,m}\sum_{j=1}^{N_\varepsilon}\pi^\varepsilon_{i,j}\Big[\big(p(S'_m)+s_{a+1}\lambda'_{m,j}\big)\Big(z_m\varepsilon_j D_1F\big(k',n^*(k',\varepsilon_j)\big)+1-\delta\Big)+s_{a+1}\zeta'_m\eta'_{m,j}\Big]=1+\frac{\lambda}{p(S)}$$

There are four terms that did not appear in (16): the multiplier on the current dividend constraint, λ, the multiplier on the future dividend constraint, λ', the multiplier on the future collateral constraint, η', and the survival probability, $s_{a+1}\equiv 1-\pi_d(a+1)$. The survival probability weights the multipliers on future constraints, which may bind with some probability. This does not matter for unconstrained firms, as, by definition, all multipliers are 0. Excess return is thus $\lambda/p(S)$, the value that an extra unit of liquidity would bring to the firm through capital investment, above the unit cost of investment.

Finally, I move to *at-risk* firms. These firms are unimpeded by their collateral today ($\eta=0$); however, they face a non-zero probability of binding future financial constraints ($\lambda'>0$). Capital investment then serves a dual purpose; it is both productive and potential collateral. These firms make a trade-off between retaining savings to use in poor future states and investing those earnings in more capital as a buffer stock for insufficient future collateral. This trade-off depends on the expected value of future capital, which is a function of $p(S'_m)$, the exact probability and degree to which future constraints bind, and their survival probability, compared to a bond's known payout of $1/q(S)$ in all states. The relationship cannot be signed; however, it can be stated that there is no guarantee of $k^*(k,b,\varepsilon,a)=k^{\mathrm{f}}(\varepsilon)$. Thus, the threat of a future binding constraint means *the same capital decision rule cannot be assigned to both unconstrained and at-risk firms.* The derivation of their decision rules and proof of this claim can be found in Appendix A1 and Lemma 1, respectively.[17] These effects still influence the decision of constrained firms; however, they are already at their borrowing limit, and cannot alter their choices further. This still increases their excess return value.

There is still the matter of issuing dividends and the unconstrained firms' debt. I follow the implementation of the *minimum savings policy* as in Khan and Thomas (2013). Firms that are currently borrowing constrained still choose $b'=\zeta k$, and thus dividends are 0. Following

[17]See online appendix for derivations of all firms' decision rules.

from before, an at-risk firm has a positive multiplier on next period's dividend non-negativity constraint, so the firm will issue zero dividends, saving as much as possible, as it values cash more than the household does in equilibrium. Unconstrained firms understand that they can issue some amount of dividends while ensuring that there is no state where they can ever become constrained. They save the minimum amount required to remain unconstrained across all possible future states. This minimum savings policy and its solution, $B^w(k, \varepsilon; S)$, for these firms are characterized as follows. Let:

$$B^w(k,\varepsilon;S) \equiv \min_{\{\varepsilon_j | \pi^{\varepsilon}_{i,j}>0 \text{ and } S'_m | \pi^{s}_{l,m}>0\}} \widetilde{B}\Big(k^{\mathrm{f}}(\varepsilon), \varepsilon_j; S'_m\Big) \qquad \text{where} \tag{19}$$

$$\widetilde{B}(k,\varepsilon;S) = z\varepsilon F\big(k, n^*(k,\varepsilon)\big) - w(S)n^*(k,\varepsilon) + q(S)\min\Big\{B^w(k,\varepsilon;S), \zeta k\Big\} - \big(k^{\mathrm{f}}(\varepsilon) - (1-\delta)k\big)$$

and finally, dividends are:

$$D(k,b,\varepsilon,a;S) = \max\Big\{0,\ x(k,b,\varepsilon,a;S) - k^{\mathrm{f}}(\varepsilon) + q(S)\min\Big\{B^w(k,\varepsilon;S), \zeta k\Big\}\Big\} \tag{20}$$

This is a conservative approach to firms' dividend decision. Given that idiosyncratic productivity follows a (bounded) Pareto distribution, the bulk of the ergodic mass concentrates at low productivity, with a long, thin spread toward high ε. Large, highly productive firms face a non-zero probability of redrawing into the low-productivity region, where the optimal scale, and hence the capacity to service debt without hitting the collateral constraint, falls sharply. However, these firms also carry more retained earnings from when they were large and can thus potentially shield themselves from such a risk.

The greater inefficiency lies in small or young firms getting a high-ε draw. Since firm growth in the model is slow, consistent with the data, these firms face high levels of excess return. These firms want to grow; they are just unable to access the necessary resources, as not even the minimum savings policy can allow them to build up sufficient retained earnings.

III. Calibration and Solution

The model's frequency is annual. The production function is Cobb-Douglas: $z\varepsilon F(k,n) = z\varepsilon k^{\alpha} n^{\nu}$. Entrant firms' capital supply is fixed at a fraction χ of the steady-state aggregate stock: $k_0 = \chi \int k\tilde{\mu}(d[k \times b \times \varepsilon \times a])$. Household utility, $U(c, 1-n^h) = \ln(c) + \psi(1-n^h)$, is the result of Hansen (1985) - Rogerson (1988) indivisible labor.

The household discount factor, β, is set to 0.962, to imply a 4 percent risk-free real interest rate, as found in Gomme, Ravikumar and Rupert (2011). The parameter governing the marginal utility of leisure, ψ, is set to 2.165, so that, in equilibrium, one-third of the household's time endowment is spent on market labor. Labor's share of output, ν, is set to 0.60 to match Cooley and Prescott (1995). I set the depreciation rate, δ, to imply an average investment-to-capital rate of 0.072, which matches the BEA Fixed Assets Accounts Tables data from 1954 to 2006.[18] χ is set to 0.243, so that the relative employment size of the average entrant is 0.260 of the average firm, matching that in the BDS from 1990 to 2006. Entrant firm leverage, $\frac{b_0}{k_0}$, is set to 0.41, to match the Ewing Marion Kauffman Firm Survey (2019). The entry rate, π_e, is set to 0.0964, in order to keep the mass of firms constant at 1, given the vector of exit rates by age, $\pi_d(a)$, in Table 1 below. Exit rates are one minus a cohort's survival rate, adjusted for growth in the number of firms to be consistent with a stationary distribution. Even though π_e is set via normalization, it is consistent with the BDS entry rate over the same horizon of about 9.94 percent.

Table 1—Exit Rates by Age

Age:	Entrant	1	2	3	4	5-9	10+	π_e
Hazard:	0.2440	0.1510	0.1251	0.1084	0.0987	0.0819	0.0589	0.0964

Note: Exit rates by age, taken from BDS (1990-2006). The BDS only reports a single firm total for ages 6-10, meaning firms age 5-9 face the implied risk.

The following parameters are jointly calibrated to match the following moments. The parameter governing capital's share of output, α, is set to 0.278 to target a capital-output ratio of 2.310. The probability of a firm maintaining its ε draw is 0.9904, to target a standard

[18] U.S. Bureau of Economic Analysis (2026*a*). To be consistent with earlier work and cited metrics, the intellectual property products are removed from these series and are not considered capital, as per the pre-1999 definition. This does not meaningfully change the calibration.

deviation of investment rates across firms of 0.337, from Cooper and Haltiwanger (2006).[19] During an ordinary financial state, the collateral constraint will be $\zeta_o = 0.864$, to target a debt-to-physical-asset ratio of 0.269, as found in the Financial Accounts (1990-2006).[20] During a credit crunch, this value changes to $\zeta_\ell = 0.4907$ in order to facilitate a peak-to-trough decrease in debt of about 32.2 percent seen from 2008Q4 to 2011Q1.[21] Finally, the model is over-identified; the number of data moments exceeds the number of parameters. The ninth, over-identifying moment is the population-weighted average of the relative size of age 0-5 firms, which is 0.368. Calibrated parameters are reported in Table 2, along with their closest related target.

Debt in the model is one-period and secured against current capital. The empirical moment chosen is loan liabilities of non-financial corporate business. Financial obligations function as loans in the model, so loans are chosen for calibration rather than debt securities.[22] Further, a credit shock in the model is disciplined by the fall in lending seen during the Great Recession, a period when bond credit *expanded*. Choosing these moments means the steady-state and crisis variables are measured against the same data.[23]

A. Size Distributions

The model captures the skewed size distribution of firms in the United States through the use of a bounded Pareto distribution for firms' idiosyncratic shock process, discretized using $N_\varepsilon = 15$ values. The discretization process is that of Jo (2025). I calculate the relative employment share of firms in the BDS, and define size bins as small (1-19 employees), medium (20-499), and large (500+). I use those relative employment share measures to define the same size cutoffs in the model distribution, and sum the mass of firms in each bin to find their population ratio.[24] I then check the resulting population share in each bin and compare

[19]This is taken from the population of incumbent continuing firms, as in Cooper and Haltiwanger (2006).

[20]Board of Governors of the Federal Reserve System (2026). This is taken as the intermediated, non-mortgage credit of non-financial corporate business, relative to non-financial tangible assets at market value.

[21]Deflated by the GDP deflator (U.S. Bureau of Economic Analysis, 2026*b*).

[22]Bonds and commercial paper are unsecured, issued almost exclusively by the largest rated firms, and underwritten on cash flows rather than collateral (Lian and Ma, 2021).

[23]Entrant leverage is measured by total debt. Entering firms rarely issue bonds or commercial paper, so this and loan measures closely coincide.

[24]Further details are in the online appendix. Decker, D'Erasmo and Moscoso Boedo (2016) are able to target the firm size distribution with permanent firm-level productivity by structuring taste shocks such that the market exposure degree of small firms is limited.

TABLE 2—PARAMETERS

	Parameter		Target	Data	Model
Panel A: Externally Assigned					
β	discount factor	$= 0.962$	real interest rate	$= 0.040$	0.040
ν	labor share	$= 0.600$	labor share	$= 0.600$	0.600
δ	depreciation	$= 0.072$	investment/capital	$= 0.072$	0.072
$\frac{b_0}{k_0}$	entrant leverage	$= 0.410$	entrant leverage	$= 0.410$	0.410
Panel B: Internally Calibrated					
ψ	leisure preference	$= 2.165$	labor hours	$= 0.333$	0.333
α	capital share	$= 0.278$	capital/output	$= 2.310$	2.320
χ	entrants' share of K	$= 0.243$	avg. n_0/N	$= 0.260$	0.260
ζ_o	collateral fraction	$= 0.864$	debt/capital	$= 0.269$	0.268
ζ_ℓ	credit crunch	$= 0.491$	decrease in debt	$= 0.322$	0.322
Panel C: Productivity Parameters					
ρ_ε	prob. retain ε	$= 0.990$	s.d. (i/k)	$= 0.337$	0.337
η_P	Pareto shape	$= 6.200$	small firm mass	$= 0.885$	0.882
ε_H	upper bound ε	$= 2.872$	medium firm mass	$= 0.112$	0.112
ε_L	lower bound ε	$= 1.000$	*normalization*		
–	–	–	pop. wtd. rel. size	$= 0.368$	0.370

Note: Panel A lists externally set parameters. Panels B and C list jointly calibrated parameters, where those in C govern the idiosyncratic shock process. Parameters are listed next to the moments they most directly influence.

ε_L is normalized to 1 and is not a chosen parameter. The relative distance between bounds is the relevant metric, but both are reported for clarity. *Large firm mass* is a residual, not a targeted moment. The model is over-identified; the "extra" moment is the *population-weighted relative size of age 0-5 firms.*

that to the data. The lower bound is normalized to $\varepsilon_L = 1.0$, and, using the population share of each firm size as calibration targets, I set the upper bound of the Pareto distribution to $\varepsilon_H = 2.872$ and the shape is set to $\eta_P = 6.20$. The resulting small, medium, and large size bins will be used as eligibility thresholds for size-based policy in Section V.B. The size bin calibration results are reported in Figure 1.

Given χ is chosen to match the average entrant's relative share of employment, the model generates an age-size distribution qualitatively comparable to what is observed in the United States for firms age 0-5, as well as matching the population-weighted mean of that group. Results for ages 1-5 are untargeted and serve as model validation. They are displayed in Table 3. Though the rate of growth measured by employment share is steeper in this model than in the data, it is still monotonic. This outcome is noteworthy given this class of model.

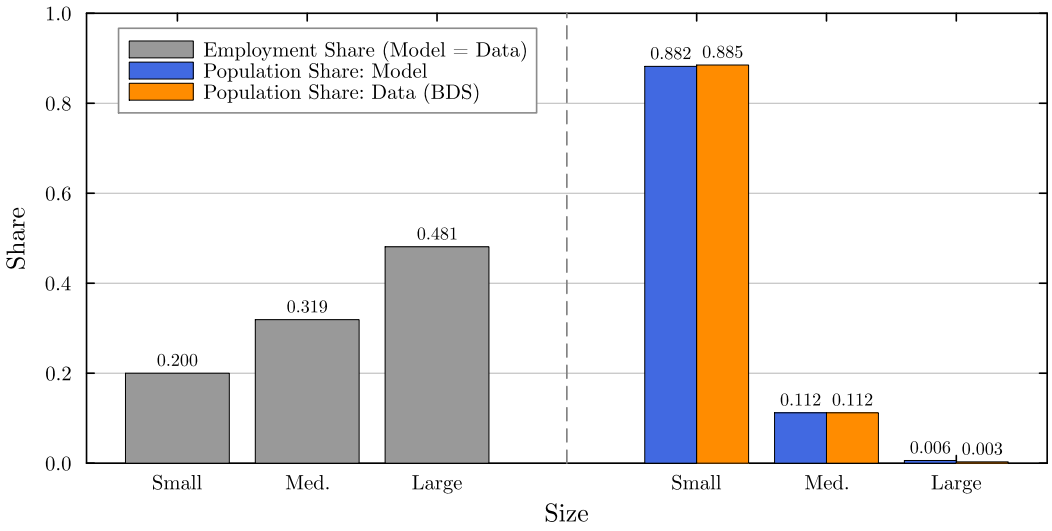


FIGURE 1. UNCONDITIONAL SIZE DISTRIBUTION

Note: Model and data employment and population shares for small, medium, and large firms.

Matching the relative size of entrants and the average for firms aged 0–5 jointly does not guarantee monotonic growth. Khan, Senga and Thomas (2024), in a related model with default risk and earnings-based borrowing limits, and obtain a distribution that is slightly decreasing at age one, though generally flat through age two, but overshoots thereafter.

TABLE 3—AGE-SIZE DISTRIBUTION OF FIRMS: BIRTH - AGE 5

Age:	Entrant	1	2	3	4	5	Pop. Wtd. Avg.
Model	0.260	0.285	0.345	0.418	0.502	0.598	0.370
Data (BDS)	0.260	0.338	0.378	0.417	0.451	0.477	0.368

Note: Relative employment share of the average firm in each age bin. The model is calibrated to match the relative share of entrants, while the remaining age bins are untargeted.

B. Solution Method

The deterministic steady state is used for calibration and is solved as follows. For a guess of p, calculate the closed-form solution of $k^{\mathrm{f}}(\varepsilon)$ for all firms and solve the minimum savings policy. Any firm that returns $D > 0$ is declared unconstrained, and its decision rules are solved. The remaining firms' decision rules are solved by simplifying the problem. All other firms choose $D = 0$ and so $b' = (k' - x)/q$; V_0 is a function of a single choice variable, k'. Though the grid is rectangular, the choice set is now a line through the space, and the

value function is concave on that line. Value function iteration, using the Howard (1960) improvement algorithm, and bilinear interpolation solve the problem and decision rules for these firms.[25] Next, the k' choice is checked against what is feasible given the financial state. If a firm is constrained by its collateral, it simply follows (17). The problem can be simplified further by realizing the absorbing age state is the only one that feeds into itself. Preceding ages already have a computed $V_0(k', b', \varepsilon', a+1)$ menu and therefore do not require iteration. Aggregates are computed and excess demand in the goods market is checked, finding a market-clearing p, which pins down w. Finally, p is updated via the Anderson and Björck (1973) root-finding method, and these steps are repeated to convergence.

The presence of aggregate uncertainty over (z, ζ) necessitates further discussion of the solution, which uses the Krusell and Smith (1998) method. I assume z is a continuous shock following a mean-zero AR(1) process in logs, $\ln z' = \rho_z \ln z + \eta'_z$ with $\eta'_z \sim N\left(0, \sigma^2_{\eta_z}\right)$, and the process is discretized to have three values using the Rouwenhorst (1995) algorithm. Using the empirical estimates of Khan and Thomas (2013), I assign $(\rho_z, \sigma_{\eta_z}) = (0.909, 0.014)$ and set the transition matrix for the two-state Markov process for ζ, where p_o and p_ℓ are the probabilities of the state persisting, as:[26]

$$\begin{bmatrix} p_o & 1-p_o \\ 1-p_\ell & p_\ell \end{bmatrix} = \begin{bmatrix} 0.9765 & 0.0235 \\ 0.3125 & 0.6875 \end{bmatrix} \tag{21}$$

To deal with the high dimensionality of μ, I assume agents do not track the full aggregate state. They know s, the total amount of capital in the economy, K, and whether or not credit markets were in a crisis state one or two periods ago, $(\vartheta_1, \vartheta_2)$.[27] Given $(s, K, \vartheta_1, \vartheta_2)$, for each of the six s realizations, agents forecast $X \in \{p, K'\}$ by:

$$\ln(X) = \beta_0 + \beta_1 \ln(K) + \beta_2 \vartheta_1 + \beta_3 \vartheta_2 \tag{22}$$

[25] V is maximized by choice of k' by placing nodes along the curve, bracketing local maxima, maximizing over those subintervals, and taking the max over each maximized interval. Though V_0 is concave, the bilinear interpolant does not inherit concavity. This helps safeguard in the event of an interpolation-caused local maximum, so that the algorithm can locate a global solution.

[26] Their estimates arise from Solow residuals on U.S. real GDP and the share of time advanced economies spend in crisis from 1945 to 2008, based on Reinhart and Rogoff (2009) data.

[27] ϑ_1, ϑ_2 are crisis dummies taking on value 1 if $\zeta_{t-1} = \zeta_\ell$ and if $\zeta_{t-2} = \zeta_\ell$, respectively.

For an initial guess of the forecasting rules, first solve $V(k, b, \varepsilon, a; s, K, \vartheta_1, \vartheta_2)$. The economy is simulated for 10,000 periods, with the first 1,000 thrown out. Each period, the period return of firm value is solved with the market clearing p, while the continuation value is built from forecasted variables, and μ' is solved. Then, for each s, regress the simulated $\ln p$ and $\ln K'$ on $\ln K$, ϑ_1, and ϑ_2. The guess is updated with the estimates, and the process repeats to convergence. Regression coefficients and statistics can be found in Appendix A4.[28]

I further simplify the problem by solving the capital choice on a one-dimensional grid of cash values when determining equilibrium outcomes. First, the value function is solved as above, with the additional aggregate states, once per update of the forecasting rules. Next period's value function is interpolated with the forecasted K', and unconstrained firms still have closed-form solutions. For firms not paying dividends, their target k' depends only on cash and (ε, a). Using the same search method as in the steady state, I compute: $k^*(x; \varepsilon, a) = \text{argmax}_{k'} \ \mathbb{E}\big[V_0\big(k', (k' - x)/q, \varepsilon', a'; S'\big)\big]$. Finally, k' is checked for feasibility, and (17) is applied accordingly. This is done to ease the computational burden of searching every (k, b, ε, a) firm each period, every iteration.

IV. Quantifying Inefficiency

A. *Steady-State Inefficiencies*

As discussed in Section II.F, there are inefficiencies present in the model before a crisis. Now with functional forms described, from the parameterization of the model, (18) becomes:

$$\frac{\beta}{p(S)} \sum_{m=1}^{N_s} \pi^s_{l,m} \sum_{j=1}^{N_\varepsilon} \pi^\varepsilon_{i,j} \Big[\big(p(S'_m) + s_{a+1}\lambda'_{m,j}\big) \Big(\alpha z_m \varepsilon_j \Big(\frac{\nu z_m \varepsilon_j}{w(S'_m)} \Big)^{\frac{\nu}{1-\nu}} k'^{\frac{\alpha+\nu-1}{1-\nu}} + 1 - \delta \Big) + s_{a+1}\zeta'_m \eta'_{m,j} \Big] = 1 + \frac{\lambda}{p(S)} \tag{23}$$

The further a firm's capital choice is from the optimal level, the more the marginal lifetime benefit of capital investment increases and the greater $\lambda/p(S)$ becomes.

The change in excess return across size groups is non-monotonic. Surprisingly, small firms, on average, have the lowest level of excess return. Even though they have the least collateral

[28] See the online appendix for simulated, HP-filtered business cycle moments comparable to the literature.

to offer, many small firms remain small on purpose; their target capital is smaller than that of medium and large firms, thus requiring less investment to reach k^*. Large firms have the most resources at their disposal, but also have the highest k^*. There is also a general equilibrium effect coming from the labor demand of large firms. The largest 0.6 percent of firms account for 48.1 percent of employment.[29] These firms drive up the price of labor, lowering optimal capital level across the economy.

Medium-size firms have the largest level of excess return, on average. These firms are more likely to have a higher idiosyncratic productivity than small firms, increasing their target capital, while also being more affected by financial conditions than large firms. Medium size is a pass-through bin; these firms ideally want to continue growing, but financial frictions slow the rate at which they can become large. Figure 2 (a) illustrates these results.

The relationship between age and excess return is more straightforward. Excess return is monotonically decreasing in age; as firms build up more retained earnings, they are more able to self-finance investment. There is also a compounding effect, especially at low levels of k, which correlates with younger firms, where increases in k loosen constraints. Even if the retained earnings are insufficient to fully self-finance, the loosening constraints increase firms' debt capacity, and thus their investment level. These effects, along with the diminishing returns to capital, create the decreasing and convex shape of Figure 2 (b).

B. Financial Crisis

I now begin the dynamic analysis of this study with a discussion of financial shocks without fiscal policy. A financial crisis in the model is an economy-wide contraction of credit, a fall from ζ_o to ζ_ℓ for four periods, followed by recovery at a rate of 31.25 percent per year, as discussed in Section III.[30] Exogenous TFP does not change in this exercise. However, the household's decision rules are still built under uncertainty; the household behaves as if z is stochastic, as it was during the simulation. This is identical to running the crisis multiple times and selecting the path where $z_t = 1, \forall t$, to isolate the impact of the credit shock.

The response to the credit crunch shares defining characteristics with the Great Recession. The fall in output is persistent as it takes time to rebuild the capital stock after the sharp

[29] In the most extreme case, the largest firm demands 633 times as much labor as the average firm.
[30] Ivashina and Scharfstein (2010) also find a large decrease in lending at the start of the Great Recession.

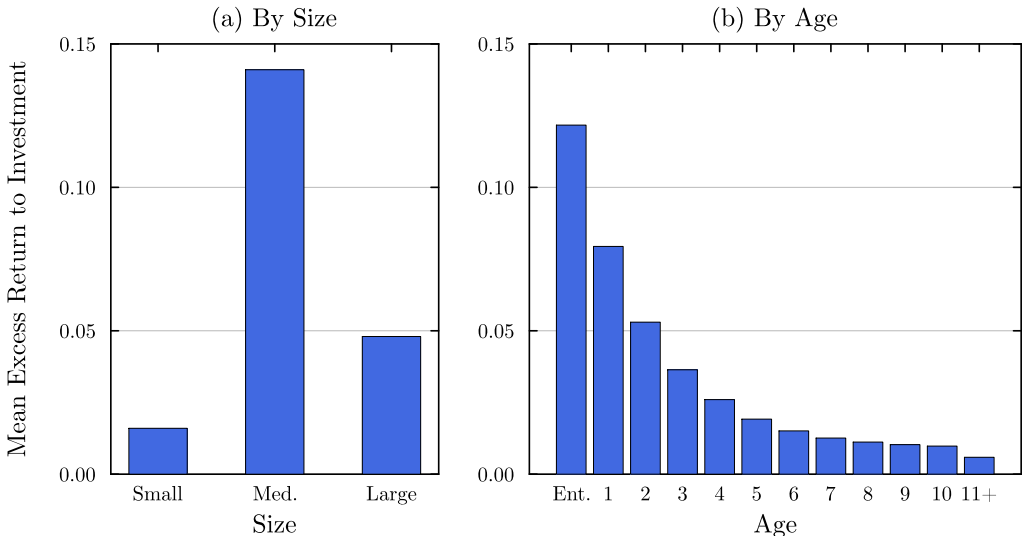


FIGURE 2. AVERAGE EXCESS RETURN TO INVESTMENT BY SIZE AND AGE

Note: The relationship between excess return and size is non-monotonic. Medium-size firms have the largest average excess return, followed by large, then small. Average excess return falls in age at a decreasing rate.

downturn in investment. The persistent decline in endogenous TFP results from changes in the distribution of firms. Financial requirements have forced the majority of firms to become smaller, without any exogenous changes to productivity, increasing misallocation of resources in the economy. There is also an initial increase in consumption, as, once the shock hits, the household foresees lower returns to saving. Figure 3 displays these dynamics.

V. Debt Relief Policy

The purpose of debt relief in the model is to relax the budget constraint of firms to enable them to afford more investment, offsetting the impact of a credit crunch. The response to the credit conditions laid out is idiosyncratic, implying the impact of different policies will be as well. Given a fixed supply of policy funds, policymakers will be concerned with the best return on the resources spent and will focus resources accordingly.

Common across all policy experiments will be the total amount of debt relief paid out, T, to ensure that each policy has the same amount of resources transferred for a fair comparison.[31] The total resource cost will be 3.5 percent of steady-state output, $\overline{Y}$, reflecting the fact

[31]It may be the case that endowing the government with more control over model variables will lead to a different T. The objective of the exercise is not to study an omnipotent policymaker, but to better understand what optimal targeting looks like given common political constraints.

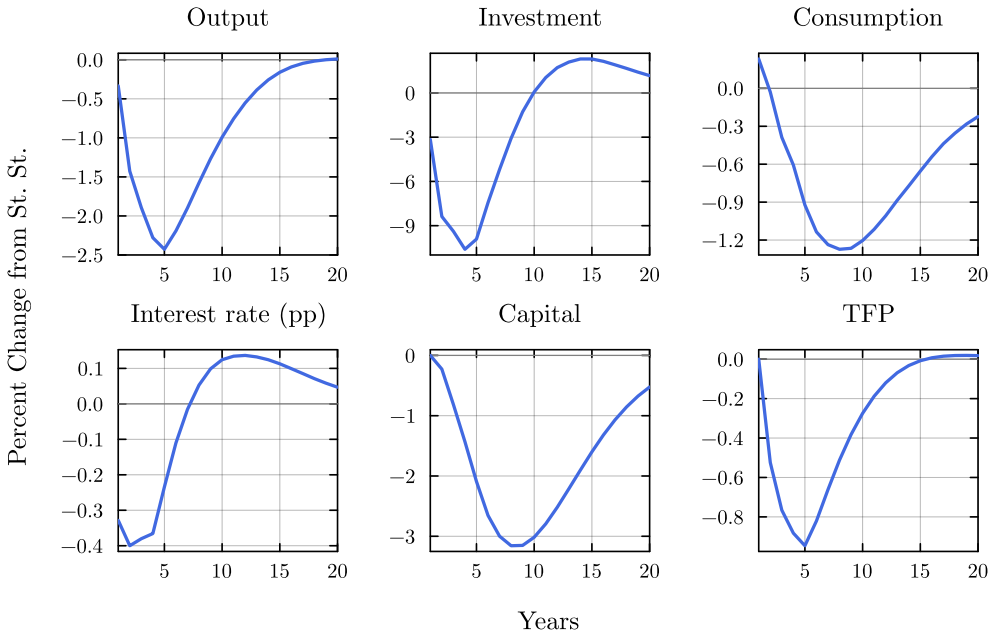


FIGURE 3. FINANCIAL CRISIS WITHOUT DEBT RELIEF

Note: Model response to an unexpected credit shock of ζ_o falling to ζ_ℓ. Shock persists to date 4 and recovers at rate 31.25 percent per year.

that the total value of PPP loan forgiveness was roughly 3.49 percent of U.S. 2020 GDP.[32] Further, policy is a response to the credit shock described above in Section IV.B and will occur unannounced on impact, so it is not something the household can forecast, nor do firms have any incentive to change their behavior in anticipation of the policy. Thus, the only variation across policy experiments is the firm-level information that the government can condition on for targeting. Table 4 summarizes this.

A. Excess Return Policy

The first policy experiment considered directly targets firm-level excess return to capital investment. This policy identifies continuing firms with the highest level of excess return and issues debt relief to them until they match the firms with the second-highest level of

[32]This is found by $\frac{F_{20}+(P_{20}\div P_{21})F_{21}}{GDP_{20}}$, where F is total forgiveness and P is the core Personal Consumption Expenditures price index. Since funds were only distributed in 2020 and 2021, this deflation method is sufficient. As of October 2022, 96 percent of all PPP loans had been forgiven (U.S. Small Business Administration, 2022).

TABLE 4—SCOPE OF INFORMATION & COMMON PARAMETERS

Panel A: Information Scope	
Policy	Scope of $\mathcal{I}$
Excess return	(k, b, ε, a), continuation
Size	(b, n)
Age	(b, a)
Panel B: Common Parameters	
Policy Cost	$0.035\overline{Y}$
Credit Shock Dates	1-4
Shock Recovery Rate ($t > 4$)	0.3125
Transfer Date	1

Note: Panel A lists the total information available to policymakers when constructing firm-level targets. Panel B lists parameters that are held constant across all debt relief policy experiments.

excess return. The policy algorithm proceeds in that manner until funds are exhausted.[33] Taking the equilibrium price sequence under the policy as given, the excess return policy is the optimal debt relief policy; it is price-taking optimal. This is proven in Appendix A2.

There is no closed-form solution for the globally optimal policy once (11) is relaxed. The interaction of the heterogeneity and financial frictions in the model means that agents have differing marginal valuations for a unit of resources, and price movements are thus not welfare neutral. The full solution is a fixed point in the relief allocation over every firm in the distribution, and their marginal effect on prices not just on impact, but over the entire horizon, both of which are functions of the household's forecast of the economy. Appendix A3 approximates the change in welfare from the price-taking solution to the full solution to be less than 0.00005 percentage points.

The intuition behind such a policy is arbitrage. The household gives up a unit of the consumption good, and one unit of debt is retired for a firm. Since the government has full information on the firms, it can ensure that these resources reach continuing firms that have the most value to gain from relieving the marginal unit of debt. Multiplying (23) by $p(S)$, the value to the household's portfolio from relieving a marginal unit of debt is $(p(S) + \lambda)$, strictly greater than the value given up by the household, $p(S)$, when the recipient firm is a

[33] Computationally, one could implement this policy as described in the text. Another way to do so would be to realize that the policy will create a level of excess return that no indebted firm will exceed. A guess of this level implies a total resource cost for the government, and this guess may be iterated on until convergence to the exogenously chosen resource cost.

continuing constrained or at-risk firm. As resources in the economy are perfectly reversible and the household holds the market portfolio, the increase in firm value by $(p(S) + \lambda)$ translates directly to an increase in value of λ for the household. Finally, starting from the highest level of excess return and working down follows a path established by diminishing returns to investment. This ensures that under the excess return policy, the next-highest λ is always reached until funds are exhausted.

The excess return policy naturally establishes a threshold for *eligible debt*. For example, it is theoretically possible for an unconstrained firm to have positive debt. This firm can reach its optimal investment level, so any debt relief resources would just be retained by the firm or passed on as dividends to shareholders. Though this just gets returned to the household, there is still the opportunity cost of those funds reaching the "wrong" firm. Thus, optimal policy would set a threshold of eligible debt such that any relief payments being made will be used for investment by the firm or, should they reach an at-risk firm, be used in expectation, which still is a net benefit for household welfare. This avoids concerns about funds not reaching their intended destination, which have been raised for the PPP (Li, 2021). The derivation of this eligible debt threshold is in the online appendix; the threshold is:

$$\max\left\{\min\left[b,\ k^{\mathrm{f}}(\varepsilon) - x(k,b,\varepsilon,a;S) - q(S)\min\left\{B^{w}(k,\varepsilon;S),\ \zeta k\right\}\right],\ 0\right\} \tag{24}$$

This expression makes sure that the firm in question has debt to be relieved in the first place, $b > 0$. Beyond that, the relief policy will only pay out to the firm until it becomes unconstrained; this is a forward-looking condition given that future multipliers may be positive. At that point the relief turns off, and the firm will not again become eligible, even if there is remaining $b > 0$. The excess return policy naturally enforces this threshold.

Figure 4 shows the impulse responses of aggregate variables to a credit shock with the excess return policy implemented. The debt relief leads to major improvements in aggregates during a crisis. The output decline at the no-policy trough date improves by 47 percent, from a 2.42 percent trough to 1.29 percent. There is a noticeable initial improvement in endogenous TFP as resources are quickly moved to their most productive location. In fact, there is even a one-period rise in investment the date after the policy is implemented, before a second fall.

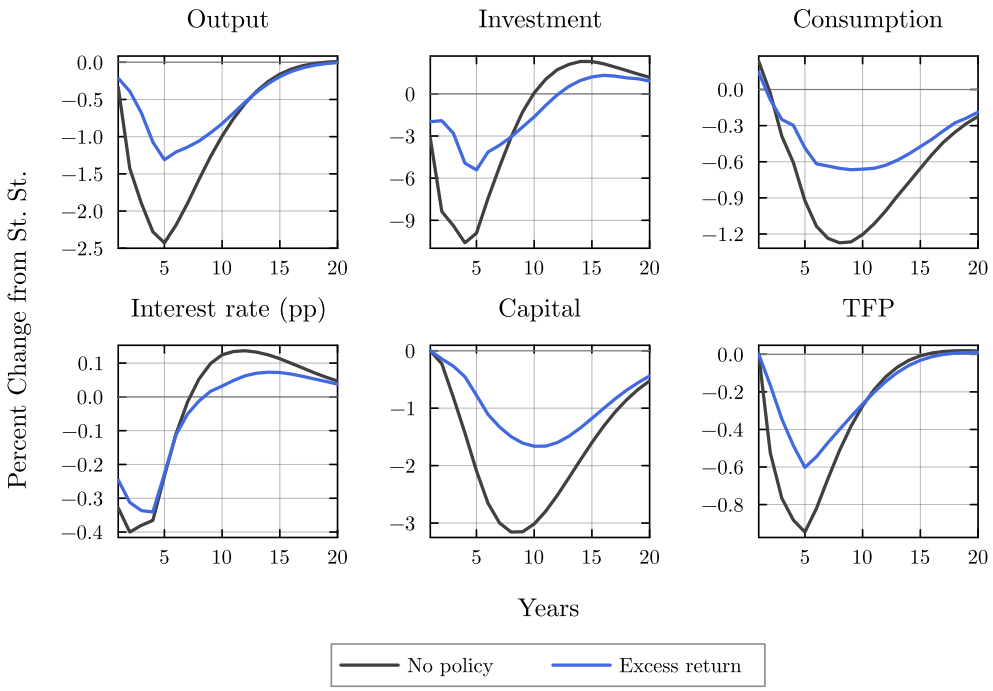


FIGURE 4. RESPONSE TO EXCESS RETURN TO INVESTMENT POLICY

Note: The excess return policy significantly dampens the effect of a crisis across all aggregates.

The policy may only last for one period, but its impact is clearly persistent, being carried on by improved firm growth. There is also a sharp uptick at date 5. This should be expected as this is the first date where it is possible to exit the crisis. This, combined with the fact that the aggregate economy is in a state more conducive to investment post-policy, allows for a relatively sharper improvement at this date.

B. Readily Observable Targets

Identifying excess return across the state space of an economy may not be feasible for policymakers. To study policies that are more likely to be implemented, I consider two readily observable policy targets: firm size and age.

SIZE-BASED TARGETS

I consider three size-based policies targeted to small, medium, and large firms. As the total size of each policy will be held constant and each targeted group is systematically different, the relief per firm must differ across policies. All indebted small firms will have 37.4 percent

of their individual debt relieved, indebted medium-size firms will have 16.8 percent, and indebted large firms will have 11.0 percent. These policies shrink the information set of the government, leading to two potential sources of opportunity cost: sending a unit of resources to a firm that does not face the highest excess return, or sending it to an exiting firm. That said, since these are such common policy targets, there is value in understanding their macroeconomic implications on a microeconomic level.

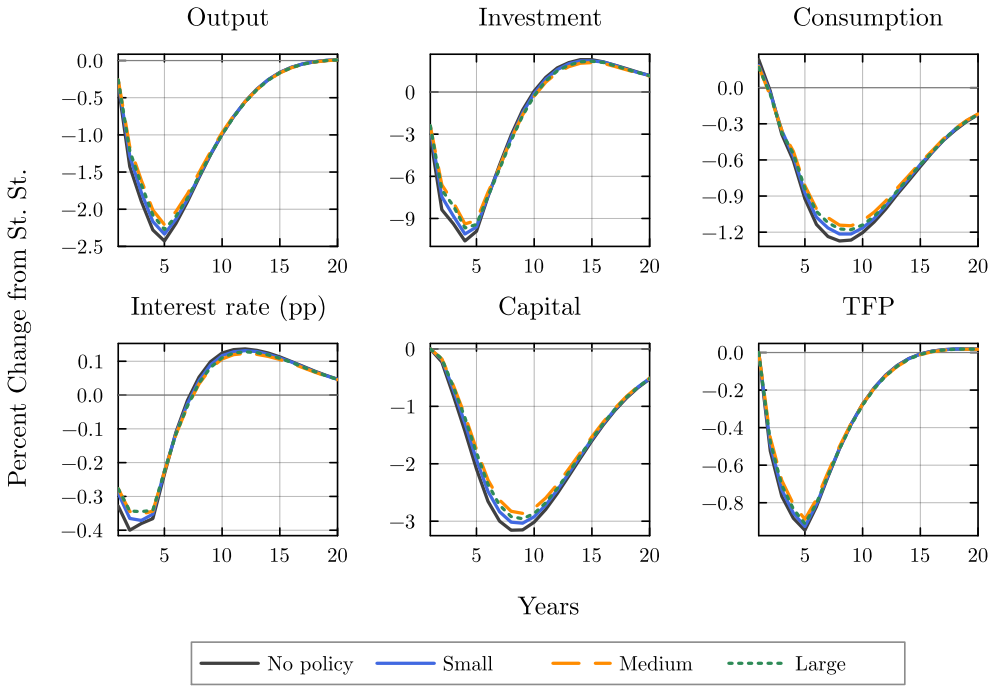


FIGURE 5. RESPONSE TO SIZE-TARGETED POLICY

Note: Across size-targeted policy, aggregates have the highest improvement under medium-size policy targets.

Figure 5 reports the response to a credit shock with a size-based policy. Targeting medium-size firms dampens the fall in aggregates more than targeting small or large firms. These firms have a larger efficient size and investment target than the average small firm, but lack the credit access to reach it. Under normal circumstances, these firms are more likely to remain medium size or become large, two groups that make up only 12 percent of firms but are responsible for 80 percent of employment and output.[34] Allowing this investment and

[34]It can be shown that for a given firm i, both output share and employment share are $\frac{(\varepsilon_i k_i^{\alpha})^{\frac{1}{1-\nu}}}{\int (\varepsilon k^{\alpha})^{\frac{1}{1-\nu}} \mu(d[k \times \varepsilon])}$.

growth to continue less impeded by the crisis provides a dynamic effect of a one-time policy, which is important during a financial shock as the largest impacts happen not on impact, but a few periods after.

Age-Based Targets

Figure 2 demonstrates age may be a good proxy for excess return. Age-based policy will target young firms (0-5 years old), middle-age firms (6-10 years old), and mature firms (11+ years old). As before, the total size of each policy will be held constant, and each group is systematically different; young firms will have 28.6 percent of their debt paid for, middle-age firms will have 20.6 percent of their debt paid, and mature firms will have 10.7 percent.

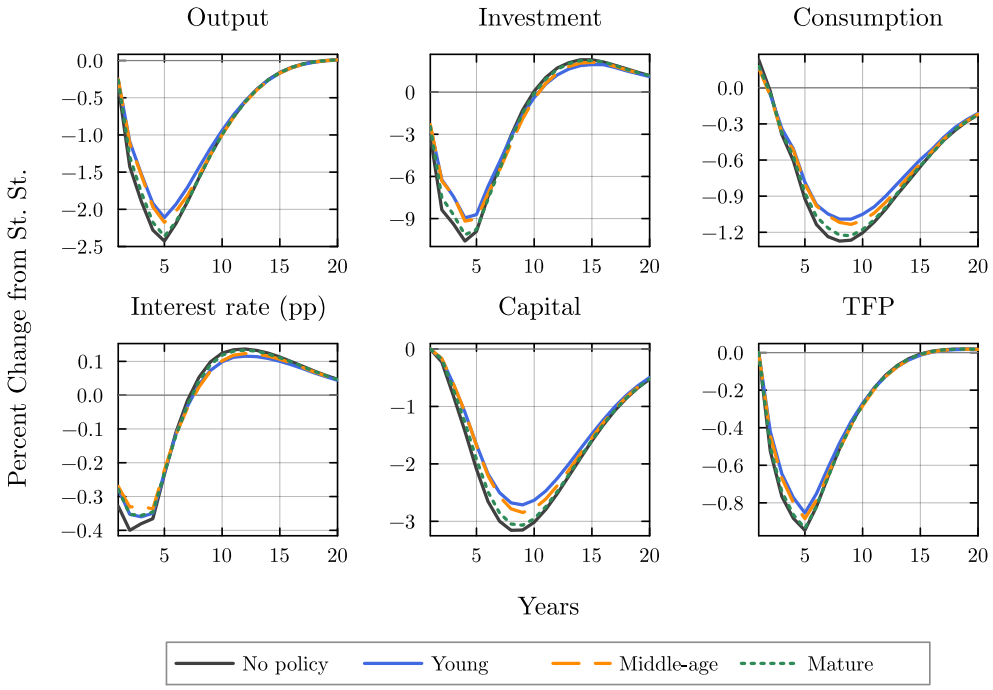


Figure 6. Response to Age-Targeted Policy

Note: Across both age- and size-targeted policy, aggregates have the highest improvement under young-firm policy targets.

Figure 6 displays the response to a credit shock under the age-targeted policies. Perhaps as expected, the young-firm policy leads to the best results. This follows from Figure 2 as young firms face higher levels of excess return to investment. Critically, this illustrates the distinction between young and small firms that is consistent with the data analyzed by Kim

et al. (2024): growth has a stronger association with young firms than small firms. Policy that allocates resources to firms with higher growth potential can more effectively dampen the impact of a crisis. Moreover, as Figure 7 shows, indebted young firms tend to be more leveraged, with a higher excess return than indebted small firms, which have the lowest leverage and excess return value across sizes.

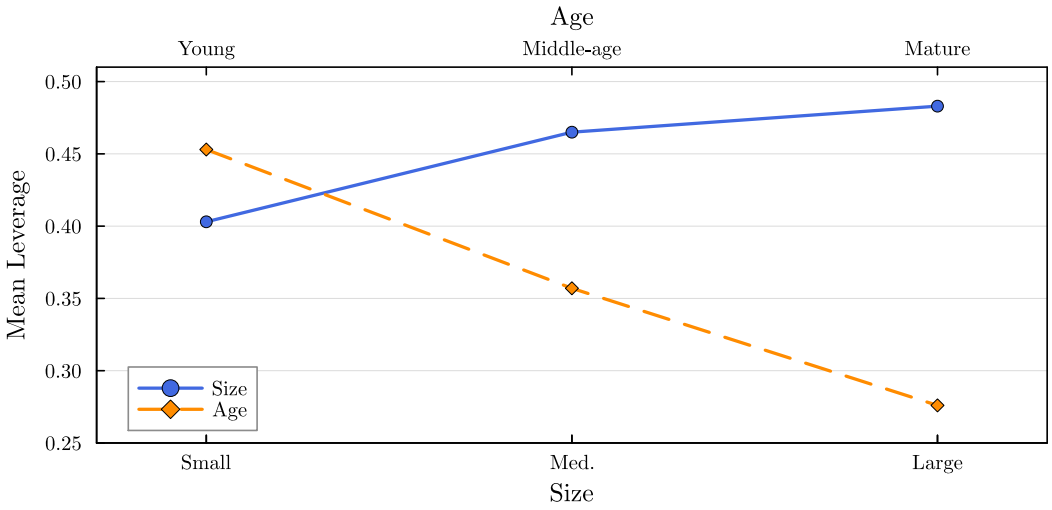


FIGURE 7. AVERAGE FIRM LEVERAGE BY AGE AND SIZE

Note: As in the data of Dinlersoz et al. (2018) for all firms, not just public firms, there is a positive relationship between firm leverage and size, but a negative relationship with age for indebted firms.

VI. Welfare Considerations

The government in this model cannot costlessly conduct policy. Though the impulse response functions generally show improvement over no policy, since the household must still forgo present consumption to fund the transfer, I calculate a welfare equivalence measure. I find the percent change in consumption given to the household in the no-policy scenario that is required to make the household indifferent between staying there and moving to one of the corresponding debt relief scenarios. That is,

$$\sum_{t=1}^{\infty} \beta^{t-1} \left[\ln \left((1 + \Delta_x) \, C_t^{\text{np}} \right) - \psi N_t^{\text{np}} \right] = \sum_{t=1}^{\infty} \beta^{t-1} \left[\ln C_t^x - \psi N_t^x \right] \tag{25}$$

where Δ_x is the percent change in *permanent* consumption required to maintain equality in (25), "np" is no policy, and x is a variable representing one of the policies in the text. The policy ranking is the same as ranking policies by minimizing cumulative output loss along the transition path. Table 5 lists the results.

TABLE 5—POLICY RANKING BY WELFARE MEASURES

Optimal		Size		Age	
Excess return	0.093%	Medium	0.018%	Young	0.026%
		Large	0.010%	Middle-age	0.016%
		Small	0.005%	Mature	0.004%

Note: Percent change in permanent no-policy consumption to make the household indifferent between no policy and each policy.

A snapshot of excess return levels does not fully explain the policy welfare ranking. In fact, it understates the importance of young firms in policy targeting. Excess return provides a value of firm growth from the marginal unit of liquidity, which diminishes with each policy dollar spent. The mean level of excess return per group provides a fine starting point, but a firm with an inelastic excess return serves as a better target than a firm with an elastic excess return. Each marginal policy dollar can provide more value when excess return does not diminish quickly, something the optimal policy naturally takes into account.

To shed light on why welfare improves more under a young firm policy than a medium-size firm policy, I approximate the elasticity of excess return, $\hat{\Theta}$. Letting $\mathrm{ER}_i(r)$ be the excess return of firm i when their state is $(k, (b-r), \varepsilon, a)$, I relieve $\Psi = 0.05$ fraction of debt in eight equal steps, re-evaluate the decisions at each step, then compute the resulting excess return. The integral evaluated by Composite Simpson's rule is:[35]

$$\hat{\Theta}_i \equiv \frac{2}{\Psi}\left(1 - \frac{\hat{\mathrm{ER}}_i}{\mathrm{ER}_i}\right), \qquad \text{where} \qquad \hat{\mathrm{ER}}_i \equiv \frac{1}{\Psi\, b_i}\int_0^{\Psi b_i} \mathrm{ER}_i(r)\, dr \tag{26}$$

On average, elasticity does not appear to change much with age. This blends two effects. As a cohort ages, an individual firm's elasticity rises as they grow, but these firms drop out of the sample once excess return hits zero. Those that remain started far from their

[35] If excess return is linear over the interval, then this is the point elasticity: $\partial \ln(\lambda)/\partial \ln(b)$.

target capital, keeping the average elasticity of the sample down. This self-financing effect is lowest for young firms since they have had the least amount of time to build retained earnings. Thus, targeting young firms has two benefits: a high average excess return *and* a low elasticity. Medium firms start out with the highest average excess return, but this diminishes at a faster rate. Conversely, the average excess return of small firms does not diminish quickly, but was very low to begin with. Table 6 lists these elasticities.

Table 6—Elasticity of Excess Return

Size		Age	
Small	0.38	Young	0.52
Medium	0.78	Middle-age	0.53
Large	1.48	Mature	0.41

Note: Approximate elasticity of excess return with respect to a fraction of relief for different firm groups.

VII. Concluding Remarks

I develop a dynamic stochastic general equilibrium model where heterogeneous firms face persistent idiosyncratic productivity shocks, financial frictions, and aggregate uncertainty. I use this model as a quantitative laboratory to study the stabilization effects of firm-targeted debt relief in a financial crisis. I employ a bounded Pareto distribution to allow the model to capture the heavily skewed size distribution of firms seen in the United States. The model also produces an age-size distribution of firms similar to that of the United States. Financial frictions limit firms' access to debt, and thus investment, providing a role for debt relief policy in a financial downturn.

Policy that can decrease and equalize the highest levels of excess return to capital investment, the expected lifetime marginal value of capital investment minus its cost, yields the best results. It allocates resources to more productive points in the distribution, cutting the fall in output nearly in half. Analyzing more commonly used targets like size and age, I find that policy targeting young firms outperforms other size and age targets, while targeting medium-size firms is the best size-based option. Providing resources to young firms allows them to invest more, speeding up their slow initial growth phase. Policy targeting medium-size firms reduces investment inefficiency in firms with relatively high growth potential, but

fewer resources available to them.

Alternative observables that correlate with excess return are a potential area for new work. Endogenizing entry and exit decisions to allow for debt relief to reduce exit costs and influence the mass of producers can be studied as a comparison to the young-firm target. I leave these ideas to future research.

Appendix

A1. Decision Rules: At-Risk Firms

All results hold at any prices that satisfy (14). Define:

$$\varphi \equiv \frac{\alpha+\nu-1}{1-\nu} < 0 \qquad c(\varepsilon;S) \equiv z\varepsilon \left(\frac{\nu z\varepsilon}{w(S)}\right)^{\frac{\nu}{1-\nu}} - w(S)\left(\frac{\nu z\varepsilon}{w(S)}\right)^{\frac{1}{1-\nu}}$$

so that operating profits, their derivative with respect to k, and cash-on-hand are:

$$\pi(k,\varepsilon;S) = c(\varepsilon;S)\,k^{\frac{\alpha}{1-\nu}}, \qquad \pi_k = \frac{\alpha}{1-\nu}\,c(\varepsilon;S)\,k^{\varphi}, \qquad x = \pi(k,\varepsilon;S) + (1-\delta)k - b$$

Let $\lambda \geq 0$ be the multiplier on the dividend non-negativity constraint, and $\eta \geq 0$ be the multiplier on the collateral constraint. Then the firm's Lagrangian is:

$$\mathcal{L} = \big(p(S)+\lambda\big)\big(x + q(S)\,b' - k'\big) + \eta\,\big(\zeta k - b'\big) + \beta\,E_{l,i}\,V_0\big(k', b', \varepsilon_j, a{+}1; S'_m\big) \tag{A1}$$

and the envelope conditions are:

$$V_k = (p+\lambda)\big(\pi_k + 1 - \delta\big) + \eta\,\zeta \qquad \text{and} \qquad V_b = -(p+\lambda) \tag{A2}$$

Iterating the Benveniste and Scheinkman (1979) result forward and substituting gives:

$$p(S) + \lambda = \beta\,E_{l,i}\Big[\big(p(S'_m) + s_{a+1}\lambda'\big)\big(\pi_k(k',\varepsilon_j;S'_m) + 1 - \delta\big) + s_{a+1}\,\zeta'_m\,\eta'\Big] \tag{A3}$$

$$q(S)\big(p(S)+\lambda\big) = \beta\,E_{l,i}\big[p(S'_m) + s_{a+1}\lambda'\big] + \eta \tag{A4}$$

Let the expected values at (k', b') of retained earnings, marginal operating profit, and additional collateral be:

$$\Lambda(k',b') \equiv E_{l,i}\big[p' + s\lambda'\big]$$

$$\mathcal{D}(k',b') \equiv \beta\,\frac{\alpha}{1-\nu}\,E_{l,i}\big[(p' + s\lambda')\,c'\big]$$

$$\bar{H}(k',b') \equiv E_{l,i}\big[s\,\zeta'\,\eta'\big]$$

Plugging $\eta = 0$ into (A4), substituting into (A3), and defining b'_{ar} as their bond choice:

$$k^* = \left[\frac{\left(\frac{\beta}{q(S)} - \beta(1-\delta) \right) \Lambda(k^*, b'_{ar}) - \beta \, \bar{H}(k^*, b'_{ar})}{\mathcal{D}(k^*, b'_{ar})} \right]^{1/\varphi} \qquad b'_{ar} = \frac{k^* - x}{q(S)} \tag{A5}$$

The presence of λ' and η' moves k^* away from the frictionless benchmark, k^{f}. There are two effects on the sign: one that pushes k^* above k^{f} and one that varies with the state.

LEMMA 1 (Relation of k^* and k^{f} for At-Risk Firms): *For an at-risk firm with state $(k, b, \varepsilon_i, a; S_l)$, let (k^*, b'_{ar}) solve (A5) and k^{f} solve (16) at the same (ε_i, S_l) and prices, with continuation multipliers evaluated at (k^*, b'_{ar}). The at-risk firm decision rule cannot be ex-ante pinned to that of an unconstrained firm; $k^* = k^{\mathrm{f}}$ is generically a knife-edge case.*

PROOF:

Since $\eta = 0$, for k^* and k^{f}, (A4) gives $p + \lambda = \beta\Lambda/q$, and substituting into (A3), dividing by $\beta\Lambda$, and using $\pi_k = \frac{\alpha}{1-\nu} \, c' \, k^{\varphi}$ gives:

$$\frac{\alpha}{1-\nu} \, (k^*)^{\varphi} \, E^{\Lambda}[c'] = a_q - \frac{\bar{H}}{\Lambda} \qquad \text{and} \qquad \frac{\alpha}{1-\nu} \, (k^{\mathrm{f}})^{\varphi} \, E^{p}[c'] = a_q$$

where $c' \equiv c(\varepsilon_j; S'_m)$, $a_q \equiv 1/q - (1-\delta) > 0$,

$$E^{\Lambda}[c'] \equiv \frac{E_{l,i}\left[(p' + s\lambda') \, c' \right]}{E_{l,i}\left[p' + s\lambda' \right]} \qquad \text{and} \qquad E^{p}[c'] \equiv \frac{E_{l,i}[p' \, c']}{E_{l,i}[p']}$$

Dividing returns the following, which defines the knife-edge case:

$$\left(\frac{k^*}{k^{\mathrm{f}}} \right)^{\varphi} = \left(1 - \frac{\bar{H}}{a_q \, \Lambda} \right) \frac{E^{p}[c']}{E^{\Lambda}[c']}$$

and $\varphi < 0$ results in:

$$k^* \geq k^{\mathrm{f}} \qquad \Longleftrightarrow \qquad E^{\Lambda}[c'] \geq \left(1 - \frac{\bar{H}}{a_q \, \Lambda} \right) E^{p}[c'] \tag{A6}$$

The claim follows as $\bar{H} \geq 0$, and $\bar{H} > 0$ when $\eta' > 0$. ■

There are two forces in (A6). The first always pushes k^* above k^{f}; capital has value as collateral, so in any case where the firm faces a positive probability of being constrained, the threshold is strictly less than $E^p[c']$.

The second force depends on future profits. The firm trades off capital and bonds, where capital pays $\pi_k + 1 - \delta$, which varies with the aggregate state, and thus $(p' + s\lambda')$, while bonds always pay $1/q$. In profitable states where funds are scarce, such as when a higher ε_j raises both c' and λ', $E^\Lambda[c'] \geq E^p[c']$. However, the opposite may occur in a crisis where $w = \psi C$ increases at the same time as λ', making bonds more attractive.

Quantitatively, these decision rules imply a 2 percent increase in k' for at-risk firms over assigning the unconstrained decision rule. This translates to a 0.4 percent increase in K under the same prices and a 0.23 percent increase in K in general equilibrium.

COROLLARY 1 (Forward-Looking Collateral Constraint): *Under a forward-looking constraint, $b' \leq \zeta k'$, at-risk and unconstrained firms still have different decision rules, despite no buffer value in k', as b'' is secured by k'' rather than by k'.*

PROOF:

Under this timing, $\eta = 0$ and $D = 0$. The first-order condition is:

$$E_{l,i}\Big[\big(p' + s\lambda'\big)\big(\pi_k(k', \varepsilon_j; S'_m) + 1 - \delta - 1/q\big)\Big] = 0$$

which k^{f} solves IFF $E^\Lambda[c'] = E^p[c']$, which will not hold if there is at least one state where the firm will no longer be at risk. ■

A2. Price-Taking Optimality of the Excess Return Policy

For each continuing, indebted firm, i, a relief schedule is $r \in \mathbf{R}^N$ with $r_i = g_i b_i$. Given $0 < T < \sum_i \mu_i b_i$, the feasible set is:

$$\mathcal{R}(T) \equiv \Big\{ r \in \mathbf{R}^N : \ 0 \leq r_i \leq b_i, \ \textstyle\sum_i \mu_i r_i = T \Big\} \tag{A7}$$

Let $\Phi_i(r) \equiv V\big(k_i, (b_i - r), \varepsilon_i, a_i; S_1\big)$. $\mathcal{W}$ is strictly increasing in wealth, which policy changes only through recipient firms, so the policy schedules can be ranked by:

$$\mathcal{V}(r) \equiv \sum_{i=1}^{N} \mu_i \, \Phi_i(r_i) \tag{A8}$$

Since the firm's constraints, and payout, are linear in $(b; D, k', b')$, and $\pi(k, \varepsilon; S)$ is concave in k, the feasible set is convex, and V and V_0 are concave in (k, b). By (A2), $-V_b = p + \lambda = p\,(1 + \mathrm{ER})$, one-sided at kinks. Thus, Φ is continuous, concave, and strictly increasing on $[0, b_i]$, with $\Phi'_{i,\pm}(r) = p\,\big(1 + \mathrm{ER}_{i,\pm}(r)\big)$, where $\mathrm{ER}_{i,\pm}(r) \geq 0$ is firm i's excess return post-relief.

A schedule $r^* \in \mathcal{R}(T)$ maximizes welfare if and only if there is an $\overline{\mathrm{ER}} \geq 0$ such that:

$$\begin{cases} \mathrm{ER}_{i,+}(r_i^*) \;\leq\; \overline{\mathrm{ER}} \;\leq\; \mathrm{ER}_{i,-}(r_i^*), & 0 < r_i^* < b_i \\ \mathrm{ER}_{i,+}(0) \;\leq\; \overline{\mathrm{ER}}, & r_i^* = 0 \\ \mathrm{ER}_{i,-}(b_i) \;\geq\; \overline{\mathrm{ER}}, & r_i^* = b_i \end{cases} \tag{A9}$$

Since $\mathcal{W}$ is strictly increasing in $\mathcal{V}$, r^* maximizes $\mathcal{W}$ if and only if it maximizes $\mathcal{V}$ on $\mathcal{R}(T)$. $\mathcal{V}$ is continuous and concave, and $\mathcal{R}(T)$ is a non-empty compact polytope, so a maximizer exists. The Karush-Kuhn-Tucker conditions are necessary and sufficient. Letting $\theta \equiv p\,(1 + \overline{\mathrm{ER}})$ be the multiplier on the budget constraint, $\Phi'_{i,+}(r_i^*) \leq \theta \leq \Phi'_{i,-}(r_i^*)$, with only the first inequality at $r_i^* = 0$ and only the second at $r_i^* = b_i$, which is (A9). Under the calibration, $\overline{\mathrm{ER}} = 0.2160$, though an unlimited budget would find the $\overline{\mathrm{ER}} = 0$ solution.

A3. Approximation of Full Optimality

Let $\Delta\mathcal{W} \equiv \mathcal{W}(r^*) - \mathcal{W}(0)$ be the welfare gain of the policy over no policy. The gain from the price-taking policy is the integral along the relief path:

$$\mathcal{G} \equiv p \sum_i \mu_i \int_0^{r_i^*} \mathrm{ER}_i(u)\, du \tag{A10}$$

so $\Delta\mathcal{W} - \mathcal{G}$ is the pecuniary term to second order. Under the calibration, this is -1.02×10^{-4} of log-permanent consumption, a negative externality to the household, as the initial fall in

consumption raises p_1 while lowering w_1 and q_1.

Policy works through creating K'. The household forecasts K_2, which is the channel through which the path of future aggregates (and thus prices) will move. To approximate the externality per value-unit of capital created, let:

$$\hat{\gamma} \equiv \frac{\Delta \mathcal{W} - \mathcal{G}}{p\,\Delta K_2} \tag{A11}$$

where $\Delta K_2 = \sum_i \mu_i \left[k'(k_i,\, b_i - r_i^*,\, \varepsilon_i,\, a_i) - k'(k_i,\, b_i,\, \varepsilon_i,\, a_i)\right]$. Under the calibration, $\hat{\gamma} = -0.0024$. For each $\gamma \in \{0, \hat{\gamma}/3, \hat{\gamma}, 3\hat{\gamma}\}$, the policy is re-solved with $\mathrm{ER}_i + \gamma\,\mathrm{MPI}_i$, where MPI_i is the marginal propensity to invest of firm i. Though $\overline{\mathrm{ER}}$ changes slightly, dropping about 2.5 percent, welfare changes by less than 0.00005 percentage points.

A4. Accuracy

Tables A1 and A2 display the results of the forecasting regressions. Although the regressions are one-step ahead forecasts, Den Haan (2010) argues that dynamic forecast errors, calculated from forecasting an entire time series without ever updating the forecast, are a more appropriate error statistic than those presented. The calculated max and mean errors over the simulation by log difference are: $e_{\max}K = 0.0160$, $e_{\text{mean}}K = 0.0021$, $e_{\max}p = 0.0015$, and $e_{\text{mean}}p = 0.0002$. For reference, these values are all below those of the Krusell and Smith (1998) stochastic-β model of 0.0202 and 0.0027, reported by Khan and Thomas (2013).

Since $e_{\max}K$ is greater than the others, I also report percentiles of the distribution of the capital forecast errors: the 50th percentile is 0.0016, the 95th is 0.0055, and the 99th is 0.0100. In the set of dates that contain the 10 largest errors, only the 7th date involves a state of financial crisis at t, or $t-1$, meaning the date the forecast was made and the preceding period are rarely ever during a financial crisis.

The autocorrelation of the one-step residual is 0.71 over consecutive periods of the same realized aggregate state, suggesting the higher $e_{\max}K$ is a consequence of the calibration, stemming from a principle similar to the self-financing of Moll (2014). Constrained firms out-grow their constraints by building retained earnings. The rate at which they do is governed by the persistence of the idiosyncratic shock process. Given this process is highly persistent, the overwhelming majority of firms take many periods to outgrow constraints. A

change in z not only changes every firm's target, but also the rate at which these retained earnings build. Two economies with the same K and z, but with varying z histories, hold different levels of liquidity and are therefore forced to invest differently. During a long, low TFP regime, earnings are too depleted for investment to match the average mean reversion rate of the predicted K', so the forecast runs ahead of the economy and the error compounds. Therefore, it may be useful to add z time dummies to the rule, in the same spirit that Khan and Thomas (2013) introduced crisis dummies. Due to computational burden, this is left for future work.

TABLE A1—CONDITIONAL FORECASTING RULES FOR AGGREGATE CAPITAL: K'

obs.	β_0	β_1	β_2	β_3	S.E.	adj. R^2	max. error
			Panel A: conditional on $s_1 = (z_1, \zeta_o)$				
2006	0.10918	0.80306	−0.00404	−0.00361	0.00085	0.99908	0.00495
	(0.00032)	(0.00055)	(0.00014)	(0.00011)			
			Panel B: conditional on $s_2 = (z_1, \zeta_\ell)$				
199	0.10344	0.80956	−0.00424	−0.00345	0.00132	0.99835	0.00368
	(0.00156)	(0.00264)	(0.00027)	(0.00025)			
			Panel C: conditional on $s_3 = (z_2, \zeta_o)$				
4170	0.13041	0.79897	−0.00430	−0.00363	0.00092	0.99892	0.00594
	(0.00027)	(0.00041)	(0.00011)	(0.00008)			
			Panel D: conditional on $s_4 = (z_2, \zeta_\ell)$				
328	0.12393	0.80576	−0.00426	−0.00355	0.00158	0.99740	0.00563
	(0.00157)	(0.00241)	(0.00024)	(0.00022)			
			Panel E: conditional on $s_5 = (z_3, \zeta_o)$				
2143	0.15184	0.79573	−0.00459	−0.00351	0.00078	0.99917	0.00517
	(0.00036)	(0.00050)	(0.00013)	(0.00011)			
			Panel F: conditional on $s_6 = (z_3, \zeta_\ell)$				
154	0.14764	0.79848	−0.00426	−0.00359	0.00123	0.99857	0.00335
	(0.00180)	(0.00253)	(0.00027)	(0.00025)			

Note: Regression: $\ln(K') = \beta_0 + \beta_1 \ln(K) + \beta_2 \vartheta_1 + \beta_3 \vartheta_2$. Standard errors in parentheses.

TABLE A2—CONDITIONAL FORECASTING RULES FOR OUTPUT VALUATION: p

obs.	β_0	β_1	β_2	β_3	S.E.	adj. R^2	max. error
			Panel A: conditional on $s_1 = (z_1, \zeta_o)$				
2006	0.68115	−0.41873	0.00142	0.00054	0.00022	0.99978	0.00136
	(0.00008)	(0.00014)	(0.00004)	(0.00003)			
			Panel B: conditional on $s_2 = (z_1, \zeta_\ell)$				
199	0.67972	−0.41974	0.00153	0.00069	0.00033	0.99961	0.00091
	(0.00039)	(0.00065)	(0.00007)	(0.00006)			
			Panel C: conditional on $s_3 = (z_2, \zeta_o)$				
4170	0.64573	−0.41342	0.00144	0.00052	0.00024	0.99973	0.00154
	(0.00007)	(0.00011)	(0.00003)	(0.00002)			
			Panel D: conditional on $s_4 = (z_2, \zeta_\ell)$				
328	0.64397	−0.41402	0.00152	0.00069	0.00039	0.99937	0.00145
	(0.00039)	(0.00060)	(0.00006)	(0.00006)			
			Panel E: conditional on $s_5 = (z_3, \zeta_o)$				
2143	0.60996	−0.40821	0.00144	0.00052	0.00021	0.99978	0.00137
	(0.00009)	(0.00013)	(0.00004)	(0.00003)			
			Panel F: conditional on $s_6 = (z_3, \zeta_\ell)$				
154	0.60822	−0.40892	0.00150	0.00068	0.00028	0.99972	0.00077
	(0.00040)	(0.00057)	(0.00006)	(0.00006)			

Note: Regression: $\ln(p) = \beta_0 + \beta_1 \ln(K) + \beta_2 \vartheta_1 + \beta_3 \vartheta_2$. Standard errors in parentheses.

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